90-Minute Hands-On Workshop

Build Your AI-Powered Digital COO & Research Assistant

The Lean Startup AI Assistant — validate an idea in under an hour, and build your own Claude Code commands & skills along the way.

Claude Code CLI · OpenCode CLI · Markdown · Git | Dr. Ibrahim Abualhaol · 18 July 2026 · ▢ QR → repo

THE PROBLEM

Your kid is struggling. The appointment is in three weeks.

What do you say tonight?

THE IDEA

Ahmad is building the answer. Alone.

😵 Founder, alone

- Re-typing the same prompts
- Weeks lost to research & docs
- Guessing instead of measuring
- Drowning in the ops grind

🤖 Founder + AI-COO

- Generates interview scripts & landing copy in minutes
- Analyzes surveys & clinician interviews instantly
- Recommends pivot-or-persevere with evidence
- Drafts SOPs, models, email sequences

LIVE DEMO

Let me show you something.

Switch to the terminal. Watch the AI read a messy CSV and reason in real time.

LIVE DEMO — RAW PROMPT

Messy CSV → market report in 30 seconds

```
claude "Read datasets/parent_survey_responses.csv (50 parents).
Analyze the data and produce a report: respondent summary, top 3
concerns, what they say they want most, willingness to pay, and key
verbatim quotes. Show the count behind every percentage.
Format as a professional markdown report."
```

The audience sees the AI read the file, reason, and write — no spreadsheet, no analyst.

LIVE DEMO — THE COMMAND 🔧

Same output. One-tenth the typing.

```
/project:analyze-survey datasets/parent_survey_responses.csv
```

Identical report — from a reusable command you'll learn to build today.

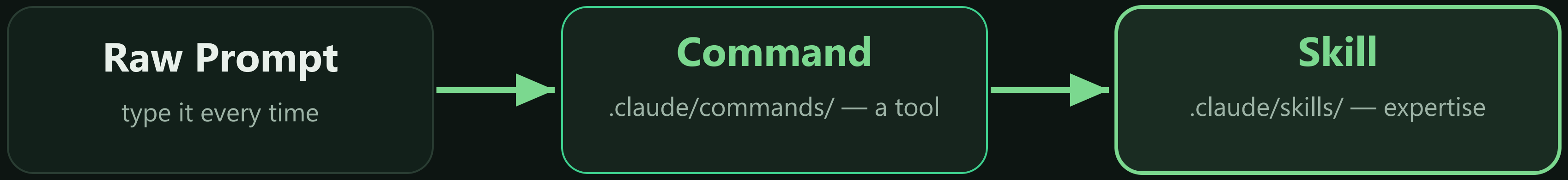
Today you won't just use prompts. You'll build tools.

THE FRAMEWORK

The Lean Startup Loop — with AI at every stage

THE PROGRESSION

Prompt → Command → Skill



Modules 1–3 build commands. Module 4 bundles them into a skill.

THE RUNNING CASE STUDY

Meet Ahmad

Healthcare data engineer building Kinsight — connecting parents and clinicians between appointments.

Stage: Pre-revenue, pre-product
Budget: $8,000 bootstrap
Team: Solo. No clinicians, no analyst, no COO.
Why him: 10 years of EHR plumbing — and a daughter who waited 5 months.

His week threads all four modules.

ROADMAP

Today — 90 minutes

Time	Module	You build 🔧
0:00–0:12	Opening — the overwhelmed founder	—
0:12–0:35	Module 1 · BUILD	`/project:hypothesis` (guided)
0:35–0:55	Module 2 · MEASURE	2 analysis commands (semi-guided)
0:55–1:15	Module 3 · LEARN	2 decision commands (independent)
1:15–1:25	Module 4 · OPERATE	`lean-startup` skill (capstone)
1:25–1:30	Closing — next steps	—

MODULE 1 · BUILD

Before you build, hypothesize

> **"We believe [segment] has [problem] and will [behavior] if we offer [solution]."**

A hypothesis is a bet you can test — specific, falsifiable, cheap to check.

MODULE 1 · BUILD

Ahmad's starting hypothesis

"We believe parents of 11-17 year olds will pay a monthly subscription for an AI risk score that tells them how their teen is really doing."

Riskiest assumption: that a *score* is what parents want. (Spoiler: the data disagrees.)

EXERCISE 1A · HANDS-ON

Generate 3 hypotheses

```
claude "I'm building a platform that helps parents and child
mental-health clinicians support a young person between appointments.
Generate 3 testable Lean Startup hypotheses in the format: 'We believe
[segment] has [problem] and will [behavior] if we [solution].'
Output a numbered markdown list with a rationale each."
```

Using your own idea? Swap the first sentence.

🔧 ANATOMY OF A COMMAND

A command is just a markdown file

```
# .claude/commands/hypothesis.md
You are a Lean Startup strategist.
Generate 3 testable hypotheses for:

$ARGUMENTS

Format each as "We believe...".

Project context:
!cat CLAUDE.md
```

1. Filename → command name

`hypothesis.md` → `/project:hypothesis`

2. $ARGUMENTS

whatever you type after the command

3. !cat — shell injection

live file contents, no copy-paste

🔧 EXERCISE 1A-CMD · GUIDED

Build your first command

```
cat > .claude/commands/hypothesis.md << 'EOF'
You are a Lean Startup strategist helping an early-stage founder.
Generate 3 testable hypotheses for the following startup idea:

$ARGUMENTS

Format each as "We believe [segment] has [problem] and will
[behavior] if we offer [solution]."

For each, also include:
- Riskiest Assumption — the belief that, if wrong, sinks it
- Cheapest Test — the fastest way to validate it
- Success Signal — the result that would confirm it

Project context:
!cat CLAUDE.md
EOF
```

Test: `/project:hypothesis a parent-clinician mental health platform`

EXERCISE 1B · HANDS-ON

Customer interview script

```
claude "Based on this hypothesis: 'parents will pay a subscription
for an AI risk score about their teen,' generate a 10-question customer
discovery interview script. Include warm-up, pain-point, solution-fit,
and willingness-to-pay questions. Format as markdown."
```

The Mom Test rule: ask about their life and past behavior — never pitch.

EXERCISE 1C · HANDS-ON

Landing page brief

```
claude "Write a landing page copy brief for a platform connecting
parents and child mental-health clinicians. Include: 1 headline (max 8
words), 1 subheadline (max 20 words), 3 value props with icons, and
1 CTA button. Target: parents of 11-17 year olds.
Tone: calm, credible, never alarmist."
```

This landing page is your Build-phase experiment — Module 2 measures it.

SHOW & TELL

What did you build?

2-3 volunteers share their output — and their first command file.

MODULE 1 · RECAP

In 23 minutes you produced...

- ✅ A tested hypothesis
- ✅ A 10-question interview script
- ✅ A landing page brief
- ✅ Your first reusable command 🔧

MODULE 2 · MEASURE

Data beats opinions.

Let AI be your research analyst.

MODULE 2 · MEASURE

Signal vs. noise

Parent survey

50 parents — what they say they need.

Clinician interviews

20 clinicians — the other side of the gap.

Competitive scan

5 platforms — what's already crowded.

New pattern this module: commands that take a file as input via `!cat $ARGUMENTS`.

MODULE 2 · MEASURE

Ahmad's data landscape

Source	What it is	Signal type
`competitors.md`	5 platforms, prices, features	Competitive intel
`parent_survey_responses.csv`	50 parent responses	Demand + willingness-to-pay
`clinician_interviews.json`	20 clinician interviews	The other side of the gap

Two populations, two files. Nobody surveyed the kids — parents and clinicians report, minors don't.

EXERCISE 2A · HANDS-ON

Competitive analysis

```
claude "Read case-study/competitors.md. Create a competitive
analysis matrix (Platform, Price, Differentiator, Target User,
Parent/Clinician Reach, Gap). Then write a 200-word landscape summary
and identify 2 underserved niches."
```

3 of 5 already ship a score

0 of 5 connect parent → clinician

🔧 EXERCISE 2A-CMD · SEMI-GUIDED

Fill in the skeleton

```
You are a competitive intelligence analyst.
Read the following competitor data:
!cat $ARGUMENTS

## Your Task
[PARTICIPANTS FILL THIS IN]

## Output Format
1. Comparison matrix: [YOU DEFINE COLUMNS]
2. 200-word landscape summary
3. 2 underserved niches
```

`!cat $ARGUMENTS` → works on any competitor file.

EXERCISE 2B · HANDS-ON

Analyze the survey (50 rows)

```
claude "Read datasets/parent_survey_responses.csv (50 parents).
Report: respondent summary, top 3 concerns, what they say they want
most, willingness to pay, key verbatim quotes. Show the count behind
every percentage. Professional markdown."
```

76% would refuse a risk score

64% just want to know what to say

56% pay $5-10, only 14% pay $20+

🔧 EXERCISE 2B-CMD · SEMI-GUIDED

Build the survey command

```
cat > .claude/commands/analyze-survey.md << 'EOF'
You are a market research analyst.
Analyze the following survey data:
!cat $ARGUMENTS

Report: respondent summary · top 3 concerns · stated top need ·
willingness to pay · verbatim quotes · 3 "So What?" insights.
[YOU ADD: rules — show the counts, flag thin data,
and say so when preference and willingness-to-pay disagree]
EOF
```

Test: `/project:analyze-survey datasets/parent_survey_responses.csv`

TECHNIQUE

The "So What?" follow-up + chaining

Iterate

```
claude "Based on that analysis, give me
exactly 3 actionable insights. For each:
evidence, implication, next step."
```

Chain 🔧

The output of `analyze-survey` becomes the input to next module's `pivot-decision`. Commands chain — mirroring the loop itself.

MODULE 2 · RECAP

In 20 minutes...

- ✅ Competitive matrix
- ✅ Survey report
- ✅ Actionable insights
- ✅ 2 reusable analysis commands 🔧

MODULE 3 · LEARN

Pivot, persevere, or zoom in?

PERSEVERE

Data supports the bet. Keep going, optimize.

PIVOT

Core thesis is wrong. Change direction.

ZOOM IN

Thesis right, lever wrong. Narrow to what works.

MODULE 3 · LEARN

Ahmad's dilemma

His bet

An AI risk score is the hero feature. Parents subscribe.

His data

76% would refuse a score about their own child. Only **12%** rank one as their top need. **64%** just want to know what to say.

The data — not his engineering instinct — makes the call.

EXERCISE 3A · HANDS-ON

Synthesize & decide

```
claude "Acting as my strategic advisor. HYPOTHESIS: parents pay a
subscription for an AI risk score. PARENTS (n=50): 76% would refuse a
score; 12% rank one as their top need; 64% want to know what to say;
56% pay $5-10, 14% pay $20+. CLINICIANS (n=20): 85% say intake is the
top pain; 70% would pay per seat; 0/20 get between-session signal.
MARKET: 3/5 rivals ship a score; 0/5 connect parent to clinician.
Should I PIVOT, PERSEVERE, or ZOOM IN? Structure:
Recommendation · Evidence · Revised Hypothesis · Risk · Confidence."
```

🔧 EXERCISE 3A-CMD · INDEPENDENT

Build it yourself — hints only

- Filename: `.claude/commands/pivot-decision.md`
- Use $ARGUMENTS for the hypothesis under review
- Use `!cat` to inject your Module-2 data files
- Output: Recommendation · Evidence · Revised Hypothesis · Risk · Confidence
- Think: what context does the AI need to decide well?

No skeleton. You've built three commands — you've got this.

AHMAD'S AI RECOMMENDATION

ZOOM IN

Revised hypothesis: clinicians will pay a per-seat licence for an intake summary built from the parent's own observations — and parents will sustain it because they get back a conversation guide, not a score.

Confidence: HIGH — three independent signals agree (parents reject it · rivals already have it · clinicians will pay for the alternative). The payer moved, and the riskiest feature is gone.

The data didn't say "wrong feature." It said don't build the diagnostic.

EXERCISE 3B · HANDS-ON

Generate a 2-week sprint plan

```
claude "Based on the zoom-in to 'clinician intake summary from
parent observations, no risk score,' create a 2-week sprint plan:
updated hypothesis, 5 prioritized tasks (solo founder), success
metrics, bootstrap budget under $1,000, risks + mitigations.
Markdown with a Gantt-style text timeline."
```

The sprint tests the scary thing: will 10 families still check in on day 14?

🔧 **EXERCISE 3B-CMD · INDEPENDENT**

Build it yourself — hints only

- Filename: `.claude/commands/sprint-plan.md`
- $ARGUMENTS = the strategic direction / pivot
- Bake in your constraints (solo, <$1,000) so you never re-type them
- Include: tasks · timeline · budget · metrics · risks

Test: `/project:sprint-plan "Pivot to transparency-focused brand"`

THE COMPLETE LOOP

Your command chain = the B-M-L loop

`hypothesis` → `competitive-analysis` `analyze-survey` → `pivot-decision` → `sprint-plan`

A complete methodology, codified — and it travels in Git.

MODULE 3 · RECAP

In 20 minutes...

- ✅ Pivot decision (evidence-based)
- ✅ 2-week sprint plan
- ✅ Revised hypothesis
- ✅ 2 decision commands — built independently! 🔧

MODULE 4 · OPERATE

Don't hire a COO. Prompt one.

MODULE 4 · OPERATE

From prompts → commands → skills

Commands = tools

You pick them up and invoke them. Single tasks.

Skills = expertise

Claude carries them and applies them automatically across a whole domain.

MODULE 4 · OPERATE

Commands vs. Skills

Dimension	Commands	Skills
Trigger	Manual — `/project:x`	Auto — context-detected
Scope	One task	A whole domain
Form	One markdown file	Folder + SKILL.md + files
Best for	Repetitive on-demand tasks	Expertise you always want
Analogy	A tool in your toolbox	A skill in your brain

EXERCISE 4A · HANDS-ON

SOP generator

```
claude "Create an SOP for onboarding a new clinic onto the
platform. Include: title + version, purpose/scope, step-by-step process
with decision points, email templates, escalation criteria, and KPIs.
Professional markdown."
```

Not this way: the safeguarding SOP. That one is written with a clinician and a lawyer — never a prompt.

🔧 EXERCISE 4B · CAPSTONE

Build the lean-startup skill

```
mkdir -p .claude/skills/lean-startup/templates

cat > .../SKILL.md << 'EOF'
---
name: lean-startup
description: Lean Startup toolkit for
  Build-Measure-Learn cycles.
---
# Lean Startup Skill
## When to Activate ...
## Methodology ...
## Principles ...
EOF
```

```
.claude/skills/lean-startup/
├── SKILL.md
├── templates/
└── examples/
```

Now Claude auto-activates your methodology whenever you open this project.

POWER TIPS

Four force multipliers

🔗 Chain

Run the whole B-M-L loop, command by command.

🧠 Context

`!cat CLAUDE.md` in every command = shared brain.

📦 Version

`git add .claude/` — your toolkit ships with the repo.

⚡ Batch

"For each company in competitors.md, generate a one-page brief and save it."

MODULE 4 · RECAP

In 10 minutes...

- ✅ A complete SOP
- ✅ The lean-startup skill 🔧
- ✅ A complete, Git-shareable AI-COO toolkit

CLOSING

Ahmad's week — before & after

Day	Produced	Built 🔧	Time
Mon	Hypotheses, script, landing	`hypothesis`	15 min
Wed	Competitive + survey, insights	2 analysis cmds	20 min
Thu	Pivot decision, sprint plan	2 decision cmds	10 min
Fri	SOPs, dashboard	SOP cmd + skill	10 min

~55 min AI-assisted vs. 2-4 weeks manual — and he killed his own best idea on Thursday.

CLOSING

Your `.claude/` folder

```
.claude/
├── commands/   ← 6 tools you invoke on demand
│   ├── hypothesis.md
│   ├── competitive-analysis.md
│   ├── analyze-survey.md
│   ├── pivot-decision.md
│   ├── sprint-plan.md
│   └── generate-sop.md
└── skills/
    └── lean-startup/   ← expertise Claude carries
```

This is your AI-COO's brain. It lives in Git.

CLOSING

5 key takeaways

- **AI is a force multiplier** — it amplifies your speed, not your judgment
- **Framework + AI = superpower** — the loop gives AI direction
- **Start tonight** — pick ONE task, prompt your AI-COO
- **Build tools, not just prompts** — commands & skills compound
- **Your .claude/ folder is your startup's operational brain**

HOMEWORK

Tonight: pick ONE task. Build a command for it.

RESOURCES

Take it with you

📦 The repo

`github.com/alhaol/ai-powered-digital-coo-research-assistant`

Commands, skill, datasets, exercises, playbook — the workshop lives in `workshop-repo/`.

📖 Keep learning

- Claude Code docs — commands & skills
- *The Lean Startup*, *The Mom Test*, *Zero to One*
- Take-home booklet PDF

Your commands & skill are local — `git push` to save them.

THANK YOU

Questions?

Don't hire a COO. Prompt one.

Dr. Ibrahim Abualhaol · AI Technical Lead · Ottawa, Canada · 18 July 2026